

M2: the falsifiable checkpoint

Which mass sets the rate of $\omega_g \rightarrow \omega_\infty$?

Abstract

We run the M0 architecture on the exactly solvable quadratic model $P(\xi) = \xi^2$, where Rosen [2] identifies the limit, and we extract the convergence *rate*, which Rosen does not. Two findings.

(1) Consistency: pass. For the quadratic model $(\mathfrak{A}, \alpha)$ is the free field of mass $m_1 = (m_0^2 + \lambda)^{1/2}$, whose locally normal ground state is unique. Hence M0 plus uniqueness reproduces Rosen's Theorem 5.2 — $\omega_g \rightarrow \omega_{m_1}$ in norm on every local algebra — with no Bogoliubov transformation, no dressing operator and no Shale condition.

(2) Rate: the strategy's conjecture was wrong by a factor 2. The exponent is governed by $2m_1$, not by m_0 and not by m_1 :

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \sim \text{poly}(d) e^{-2m_1 d}, \quad d = \text{dist}(O, \{g < 1\}).$$

The mass is the *physical* one because the resolvent kernels decay at the local Agmon rate, which is m_1 inside $\{g = 1\}$; the factor 2 is a *round trip* $O \rightarrow \text{supp}(1 - g) \rightarrow O$ forced by the exact resolvent identity. Confirmed numerically to $\sim 1\%$ over six decades (§4).

1 The model and the two questions

Take $P(\xi) = \xi^2$, i.e.

$$H(g) = H_0 + \frac{\lambda}{2} \int_{\mathbb{R}} g(x) : \varphi_0(x)^2 : dx - E(g), \quad \lambda > 0,$$

and set

$$A := -\Delta + m_0^2 + \lambda g, \quad B := -\Delta + m_1^2, \quad m_1^2 := m_0^2 + \lambda,$$

so that $\mu_g := A^{1/2}$ and $\mu_1 := B^{1/2}$ are the one-particle energies. Both ω_g and the candidate limit ω_∞ are quasi-free:

$$\omega_g(W(f, h)) = \exp\left(-\frac{1}{4}[\langle f, \mu_g^{-1} f \rangle + \langle h, \mu_g h \rangle]\right), \quad (1)$$

and likewise for ω_∞ with μ_1 . Everything therefore reduces to the two operator differences

$$\Delta_\varphi := \mu_g^{-1} - \mu_1^{-1}, \quad \Delta_\pi := \mu_g - \mu_1. \quad (2)$$

Fix a bounded open O and let I be an interval with $g \equiv 1$ on I and $O \subset I$; put $d := \text{dist}(O, \mathbb{R} \setminus I)$.

2 Consistency: the architecture reproves Rosen

Lemma 2.1. *For the quadratic model, the Glimm–Jaffe automorphism group α of $\mathfrak{A}$ is the free time evolution of mass m_1 .*

Proof. Let $A \in \mathfrak{A}(O)$ and $t \in \mathbb{R}$, and pick $g \in \mathcal{G}$ with $g \equiv 1$ on $O_{|t|}$. By Fact 1.3(ii) of [1], $\alpha_t(A) = e^{itH(g)} A e^{-itH(g)}$. The Heisenberg equations for $H(g)$ are the linear field equations $\partial_t^2 \varphi = \Delta \varphi - (m_0^2 + \lambda g) \varphi$, whose solution on $O_{|t|}$ depends, by finite propagation speed, only on the coefficients in $O_{|t|}$, where $m_0^2 + \lambda g \equiv m_1^2$. Hence α_t restricted to $\mathfrak{A}(O)$ is the free mass- m_1 evolution. As O and t were arbitrary and $\bigcup_O \mathfrak{A}(O)$ is norm-dense, the claim follows. $\square$

Lemma 2.2. *The free field of mass $m_1 > 0$ has a unique locally normal ground state, namely the quasi-free state ω_{m_1} with covariances $(2\mu_1)^{-1}$ and $\mu_1/2$.*

Proof sketch. Let ω be a ground state. In the GNS representation the generator $H_\omega \geq 0$ annihilates Ω_ω , and the free dynamics acts on the one-particle test-function space by the symplectic group generated by μ_1 . Positivity of H_ω forces the two-point function to be the μ_1 -vacuum one, exactly as in the classical argument for the free field; local normality excludes the non-regular ground states carried by non-normal representations. Uniqueness of the one-particle covariance plus quasi-freeness of ω_{m_1} then gives $\omega = \omega_{m_1}$. $\square$

Corollary 2.3 (consistency check: **pass**). *Assume the local number bound (H2) of [1]. Then, in the quadratic model, $\omega_g \rightarrow \omega_{m_1}$ in norm on every $\mathfrak{A}(O)$.*

Proof. Combine [1, Thm. 4.2, Cor. 4.4] with Lemmas 2.1 and 2.2. $\square$

This is exactly Rosen’s Theorem 5.2 [2, §5]. The point is the route: Rosen identifies the limit through the explicit dressing transformation $T_g = e^{iL}\Gamma(u)$ and the Shale condition $\ln p \in \text{HS}$, none of which appears above. The M0 architecture obtains the same conclusion from the exact ground-state identity plus uniqueness, i.e. from precisely the two ingredients that are also available — in principle — for φ_2^4 .

3 The rate

3.1 The exact resolvent identity, and why a round trip appears

Since $A - B = -\lambda(1 - g)$ and $(A + s)^{-1} - (B + s)^{-1} = -(A + s)^{-1}(A - B)(B + s)^{-1}$, the standard representations $S^{-1/2} = \frac{1}{\pi} \int_0^\infty s^{-1/2} (S + s)^{-1} ds$ and $S^{1/2} = \frac{1}{\pi} \int_0^\infty s^{-1/2} S (S + s)^{-1} ds$ give the *exact* identities

$$\Delta_\varphi = \frac{\lambda}{\pi} \int_0^\infty s^{-1/2} (A + s)^{-1} (1 - g) (B + s)^{-1} ds \geq 0, \quad (3)$$

$$\Delta_\pi = -\frac{\lambda}{\pi} \int_0^\infty s^{+1/2} (B + s)^{-1} (1 - g) (A + s)^{-1} ds \leq 0. \quad (4)$$

These are not perturbative expansions: they are identities. The signs agree with $\mu_g \leq \mu_1$ and are confirmed numerically in §4.

The structural consequence is immediate and is the whole content of M2. The factor $1 - g$ is supported in $\mathbb{R} \setminus I$, at distance $\geq d$ from O . Hence the kernel of $P_O \Delta_\varphi P_O$ is a sum over paths that leave O , reach $\text{supp}(1 - g)$, and return: the exponential cost is paid *twice*.

3.2 Which mass

Lemma 3.1 (local Agmon decay). *For $s \geq 0$ let $\kappa_j(s) := (m_j^2 + s)^{1/2}$. Then*

$$(B + s)^{-1}(x, y) = \frac{1}{2\kappa_1} e^{-\kappa_1|x-y|} \quad (\text{exactly, in one dimension}),$$

and there is C independent of s, g with

$$|(A + s)^{-1}(x, z)| \leq \frac{C}{\kappa_0} \exp\left(-\int_x^z (m_0^2 + \lambda g(u) + s)^{1/2} du\right).$$

The second bound is the standard Agmon/Combes–Thomas estimate; the point is that the decay rate is the *local* one. For $x \in O$ and $z \notin I$ the path crosses I , where $m_0^2 + \lambda g \equiv m_1^2$, so

$$\int_x^z (m_0^2 + \lambda g + s)^{1/2} du \geq \kappa_1(s) d + \kappa_0(s) (\text{dist}(z, I)).$$

Proposition 3.2. *There is $C = C(m_0, \lambda, |O|)$ such that for all $d \geq 1$,*

$$\|P_O \Delta_\varphi P_O\| \leq C d^{-1/2} e^{-2m_1 d}, \quad \|P_O \Delta_\pi P_O\| \leq C d^{-3/2} e^{-2m_1 d}.$$

Moreover the exponent is sharp: by (1), $\omega_g(\varphi_0(f)^2) - \omega_\infty(\varphi_0(f)^2) = \frac{1}{2} \langle f, \Delta_\varphi f \rangle$ for $\text{supp } f \subset O$, so $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\|$ is bounded below by a constant times $\|P_O \Delta_\varphi P_O\|$ as well as above.

Proof sketch. Insert Lemma 3.1 into (3). The z -integral over $\mathbb{R} \setminus I$ gives a factor $\leq 2e^{-\kappa_1 d}/(\kappa_0 + \kappa_1)$, so the two exponentials combine to $e^{-2\kappa_1(s)d}$. Substituting $\kappa = \kappa_1(s)$, $s = \kappa^2 - m_1^2$, $ds = 2\kappa d\kappa$,

$$\int_0^\infty s^{-1/2} e^{-2\kappa_1(s)d} \frac{ds}{\kappa_1^2(\kappa_0 + \kappa_1)} \asymp \int_{m_1}^\infty \frac{2\kappa e^{-2\kappa d}}{(\kappa^2 - m_1^2)^{1/2} \kappa^2(\kappa_0 + \kappa)} d\kappa \asymp C d^{-1/2} e^{-2m_1 d},$$

the last step by Watson’s lemma at the endpoint $\kappa = m_1$, where the integrand has an inverse square-root singularity. For (4) the weight $s^{+1/2}$ replaces $s^{-1/2}$, turning $(\kappa^2 - m_1^2)^{-1/2}$ into $(\kappa^2 - m_1^2)^{+1/2}$ and hence $d^{-1/2}$ into $d^{-3/2}$. $\square$

Remark 3.3 (what M2 falsifies). The strategy note conjectured $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \lesssim e^{-m d}$ with m the mass gap. Proposition 3.2 says the correct statement carries a factor 2 in the exponent. Both features generalise beyond the quadratic model: the round trip is forced by the resolvent identity, i.e. by the fact that the perturbation is *not* between O and the observable but between O and the region where the two Hamiltonians differ; and the relevant mass is the local one inside $\{g = 1\}$, which is the physical mass, not the bare mass m_0 . Any quasi-adiabatic argument at Milestone M4 must reproduce $2m$, not m .

4 Numerical confirmation

Script: `m2_rate.py`; output: `m2_rate.out`. Parameters $m_0 = 1$, $\lambda = 3$, hence $m_1 = 2$ and the four candidate exponents are $m_0 = 1$, $2m_0 = 2$, $m_1 = 2$, $2m_1 = 4$. Sharp cutoff $g = \mathbf{1}_{[-\ell, \ell]}$, box $[-25, 25]$, spacing $h = 0.025$ (and $h = 0.0166$ for a refinement check), $O = \{0\}$ so $d = \ell$. Kernels evaluated at the origin; one eigendecomposition per ℓ .

	fitted exponent k	prefactor power p	max. residual
Δ_φ , joint fit $\log y = c - kd - p \log d$	4.0343	0.288	$1.1 \cdot 10^{-3}$
Δ_π , joint fit	4.0767	1.019	$4.0 \cdot 10^{-3}$
prediction	$2m_1 = 4$	$1/2, 3/2$	—

Prefactor-free evidence: the local slopes $k(d) = -d \log |y|/dd$, which for $|y| \sim d^{-p}e^{-kd}$ satisfy $k(d) = k + p/d$, decrease monotonically over $d \in [1, 4]$ from 4.285 to 4.105 (Δ_φ) and from 4.965 to 4.322 (Δ_π); subtracting p/d with the fitted p gives 4.031 and 4.060. Refining h by $1.5\times$ moves the Δ_φ exponent by < 0.03 . Both sign predictions of (3)–(4) hold at every data point.

Verdict. $k = 4.03 \pm 0.05$ against candidates $\{1, 2, 2, 4\}$: the exponent is $2m_1$, decisively. The polynomial prefactor powers come out 0.29 and 1.02 against the predicted 0.5 and 1.5; over $d \in [1, 4]$ the asymptotic expansion has relative corrections of order $1/d$, so these are consistent in sign and magnitude but are *not* sharply confirmed. The exponent, which is what M2 was designed to test, does not depend on them.

5 Consequences for the programme

1. **The architecture is sound.** Corollary 2.3 recovers a known theorem from M0 plus uniqueness, using no structure special to the quadratic case beyond the uniqueness input itself. This is the proof-of-concept the strategy needed.
2. **Correct the rate in the strategy:** $e^{-m d} \rightsquigarrow e^{-2m d}$ throughout §6, and correspondingly in the target statement of Theorem A.
3. **Rosen’s estimates are not sharp for this question.** His Lemma 5.1 carries $e^{-m_0|j|}$ because it bounds the *free, mass- m_0* kernel; it is used only to prove $\|(\mu_g - \mu_1)\xi\| \rightarrow 0$ qualitatively. The sharp local rate involves m_1 and the factor 2. This is consistent with Rosen never stating a rate.
4. **A target for M4.** Any quasi-adiabatic/LPPL argument must produce $2m$; if it produces m , it is lossy by a factor 2 in the exponent, and if it produces m_0 it is wrong.

References

- [1] *Limit points of the finite-volume ground states of $P(\varphi)_2$ are ground states of the quasi-local dynamics*, Milestone M0, `../m0/m0.tex`.
- [2] L. Rosen, *Renormalization of the Hilbert space in the mass shift model*, J. Math. Phys. **13** (1972) 918–927.